```
Amarm 50219 AgSp1.1 glycine rm1 1         PGTTPGTITGPDGKPT.KFIVPYGAFTTPGSIPGPDGRPIH.VEPAGPGTTPGAETGPDGKVNK......IYLPTTPK..G.PGG   74
Lcorn 18104 AgSp1.1 glycine rm1 1         PGTTPGTVTGPDGKPI.KFIVPTGAFTTPGSIPGPDGKPIH.VEPAGPGTTPGAQTGPDGKINK......LVLPTTPK..APQGP   75
Ncruc 58532and13532 AgSp1 glycine rm 1   PGTTPGTVTGPDGKPT.KFVVPLGAFTTPGSIPGPDGQPIH.VEPAGPGTTPGAETGPDGKVN......KIYLPTTP...KGPGG   74
Mlaby 7048 AgSp1 glycine rm1a 1          ...............................................PGTTPGAETGPDGNVDK......IILPTTPVRPEPQPQ   32
Mhut 1271 AgSp1 glycine rm1 1            PGTTPGTITGPDGKPI.QFIVPYGAFSTPGSIPGPDGIPIL.VEPAESGKTPGIETGPDGKIYK......IYLPIIQK..GPDKH   75
Aarg AgSp1 FC511 glycine rm1 1           PGTTPGTITGPDGKPT.KFIIPLGAFTTPGSIPGPDGRPIH.VQPAGPGTTPGAETGADGKINK......IIVPTTPKS......   71
Atri AgSp1 genomic glycine rm1 1         PGTTPGTITGPDGKPT.KFIIPLGAFTTPGSIPGPDGRPIH.VQPAGPGTTPGAETGPDGKINK......IIVPTT.........   68
Msag 18528 AgSp1.2 glycine rm 1          PGTTPGTVTGPDGKP.R....................................................................   16
Varen 6869 AgSp1.2 glycine rm1 1         PGTTPGTITRHDGKPK.EFIIPIGAFTTPGSFPGPDGRPIY.VEPAGPGTTPGAITGPDGKIYK......IIVPTTQK.......   70
Varen 6869 AgSp1.1 glycine rm1 1         .................KFIIPNGAFRTPGSVPGPDGEPIS.VEPAGPGTTPGTETGPDGKIRK......VVIPTTPK..GPHGP   59
Amarm 50219 AgSp1.1 glycine rm1 2         PGTTPGTVTGPDGKPT.KFIVPTGAFTTPGSIPGPDGRPIH.VEPAGPGTTPGAETGPDGKINK......IYLPSTPK..GPGGP   75
Ncruc 58532and13532 AgSp1 glycine rm 2   PGTTPGTVTGPDGKPT.KFVVPLGAFTTPGSIPGPDGQPIH.VEPAGPGTTPGAETGPDGKVN......KIYLPTTP...KGPGG   74
Mlaby 7048 AgSp1 glycine rm1a 2          PGTTPGTVTGPDGKPS.KYIVPLGAYTTPGSIPGPDGKPIH.VQPAGPGTTPGTITDSQGNVDK......IILPTTPK..GQQ..   73
Mhut 1271 AgSp1 glycine rm1 2            PGTTPGTITGPDGRPT.KFIVPYGAFTTPGTIPGPDGKPIQ.VEPAGPGTTPGVERGDDGKVYK......IYLPLMHI..S.PGE   74
Aarg AgSp1 FC511 glycine rm1 2           PGTTPGTVTGPDGKPK.KFVVPKGAFTTPGSIPGPDGKPIH.VEPAGPGTTPGAQTGPDGKINK......LVVPTTTTPKGPVGP   77
Atri AgSp1 genomic glycine rm1 2         PGTTPGTVTGPDGKPK.KFVLPKGAFTTPGSIPGPDGKPIH.VQPAGPGTTPGAQTGPDGKINK......LVVPTTTTPKGPVGP   77
Varen 6869 AgSp1.2 glycine rm1 2         PGTTPGTITGSDGKPT.KFIIPMGAFTTPGSMPGPDGRPIR.VEPAGPGTTPGVVTRHDGKIYK......IIVPAMQMGSDE...   74
Varen 6869 AgSp1.1 glycine rm1 2         PGTTPGTVTGRDGKPR.KFIVPNGAFTTPGSFPGPDGEPIS.VEPAGPGTTPGIETDSDGKIKK......VVVPSTPK..GPEGP   75
Amarm 50219 AgSp1.1 glycine rm1 3         .....................................................................................    0
Ncruc 58532and13532 AgSp1 glycine rm 3   PGTTPGTVTGPDGKPT.KFVVPLGAFTTPGSIPGPDGQPIH.VEPAGPGTTPGAETGPDGKVN......KIYLPTTP...KGPGG   74
Mhut 1271 AgSp1 glycine rm1 3            PGTTPGTITGPDGKPT.KFIVPYGAFTTPGSIPGPDGKPIH.VEPAGPGTTPGVKRGPDGKVYK......IYLPTTPK..G.PGG   74
Aarg AgSp1 FC511 glycine rm1 3           PGTTPGTVTGSDGKPK.KFIVPK..............................................................   22
Varen 6869 AgSp1.2 glycine rm1 3         PGTTPGTVTGK..........................................................................   11
Ncruc 58532and13532 AgSp1 glycine rm 4   PGTTPGTVTGPDGKPT.KFVVPLGAFTTPGSIPGPDGQPIH.VEPAGPGTTPGAETGPDGKVN......KIYLPTTP...KGPGG   74
Mlaby 7048 AgSp1 glycine rm1a 4          PGTTPGTVTGPDGKPT.KFIVPTGAFTTPGSIPGPDGKPIR.VEPAGPGTTPGAETGPDGKINK......LVVPTTPK..E.P..   72
Mhut 1271 AgSp1 glycine rm1 4            PGTTPGTITGSDGRPI.QFIIPYGAYSTPGSIPGPDGTPIH.VEPAGPGTTPGIETGPDGKVYK......IYLPTTPK..G.PGG   74
Ncruc 58532and13532 AgSp1 glycine rm 5   PGTTPGTVTGPDGKPT.KFVVPLGAFTTPGSIPGPDGQPIH.VEPAGPGTTPGAETGPDGKVN......KIYLPTTP...KGPGG   74
Mhut 1271 AgSp1 glycine rm1 5            PGTTPGTITGSDGRPI.KFIIPYGAYSTPGSIPGPDGTPIH.VEPAGPGTTPGIETGPDGKVYK......IYLPTRPK......R   71
Ncruc 58532and13532 AgSp1 glycine rm 6   PGTTPGTVTGPDGKPT.KFVVPLGAFTTPGSIPGPDGQPIH.VEPAGPGTTPGVQTGPDGK........................   59
Mlaby 7048 AgSp1 glycine rm1a 6          PGTTPGVVTGPDGQPV.QFIVPQGAFTTPGSIPGPNGKRIH.VKPAGP.....................................   46
Mhut 1271 AgSp1 glycine rm1 6            PGTTPGTITGSDGRPI.KFIIPYGAYSTPGSIPGPDGTPIH.VEPAGPGTTPGVKTGPDGKVYK......IYLPTTPK..G.PGG   74
Mhut 1271 AgSp1 glycine rm1 7            PGTTPGTLTGSDGRPI.KFIIPYGAFSTPGSIPGPDGTPIH.VEPAGPGTTPGVKTGPDGKVYK......IYLPTTPK..G.PGG   74
Amarm 50219 AgSp1.1 glycine rm1 8         PGTTPGTVTGPDGKPI.KFIVPTGAFTTPGSIPGPDGRPIH.VEPAGPGTTPGAETGPDGKINK......IYLPSTPK..G.PGG   74
Mhut 1271 AgSp1 glycine rm1 8            PGSTPGTITGPDGRPT.QFIIPFGAFTTPGSIPGPDGIPIH.VEPAG......................................   45
Varen 6869 AgSp1.2 glycine rm1 10        ......................................................................IIVPTIHM.......    8
Varen 6869 AgSp1.1 glycine rm1 11        ........................................................GPEG.........................    4
Varen 6869 AgSp1.2 glycine rm1 11        PGTTPGTVTGKDGKPT.KFIIPIGAFTTPGSIPGPDGRPIR.VEPAGPGTTPGVVTRPDGKIYK......IIVPTIHMGSDE...   74
Varen 6869 AgSp1.2 glycine rm1 12        PGTTPGTVTGNDGKPT.KFIIPIGAFTTPGSIPGPDGRPIR.VEPAGPGTTPGVVTRPDGKIYK......IIVPTIHMGSDE...   74
Aarg AgSp1 FC511 glycine rm1 13          .......................GAFTTPGSIPGPDGKPIH.VEPAGPGTTPGAQTGPDGKINK......IVVPTTTTSKGPVGP   55
Varen 6869 AgSp1.2 glycine rm1 13        PGTTPGTVTGKDGKPT.KFIIPIGAFTTPGSIPGPDGRPIR.VEPAGPGTTPGVVTRPDGKIYK......IIVPTIHMGSDE...   74
Varen 6869 AgSp1.2 glycine rm1 14        PGTTPGTVTGKDGKPI.KFIIPIGAFTTPGSIPGPDGRPIR.VEPAG......................................   45
Mlaby 7048 AgSp1 glycine rm1a 15         PGTTPGVVTGPDGQPV.QFIVPQGAFTTPGSIPGPDGKPVH.VRPAGPGTTPGAKNPLPVTQSD..GQPIQIVPA........GP   73
Aarg AgSp1 FC511 glycine rm1 15          PGTTPGTVTGPDGKPK.KFVVPQGAFTTPGSIPGPDGNPIP.VEPAGPGTTPGTQTGPDGKINK.....................   62
Msag 18528 AgSp1.2 glycine rm 16         .....................................................................................    0
Lcorn 18104 AgSp1.1 glycine rm1 37        .GTTPGTVTGPDGKPT.KFIVPKGAFTTPGSIPGPDGKPIH.VEPAGPGTTPGAETGPDGKINK......LVVPTTPK..GPQGP   74
Lcorn 18104 AgSp1.1 glycine rm1 38        PGTTPGTVTGPDGKPN.KFIVPKGAFTTPGSIPGPDGKPIH.VEPAGPGTTPGAQTGPDGKINK......LVVPTTPK..GPQGS   75
Lcorn 18104 AgSp1.1 glycine rm1 39        PGTTPGTVTGPDGKPT.KFIVPKGAFTTPGSIPGPDGKPIH.VEPAG......................................   45
Atri AgSp1 genomic glycine rm1 47        .....................................................................................    0
Msag 18528 AgSp1.2 glycine rm 112        .....................................................................................    0
Msag 18528 AgSp1.2 glycine rm 113        PGTTPGTVTGPDGKP.KKFVVPKGAFTTPGSFSGPDGQPIH.VEPAGPGTTPGVQTGPDGKINK......VVVPTTTK..GPQGP   75
Msag 18528 AgSp1.2 glycine rm 114        PGTTPGTVTGPDGKP.KKFVVPKGAFTTPGSFSGPDGQPIH.VEPAGPGTTPGVQTGPDGKINK......VVVPTTTK.......   70
Msag 18528 AgSp1.2 glycine rm 115        ..........SDGKP.QKFVIPKGAFTTPGFFPGPDGQPIH.VEPAK......................................   35
consensus                                 ***************  ***** ************** *** ***********  ******* *      * ****
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Amarm 50219 AgSp1.1 glycine rm1 1          ..........PGG............................................QTTPTPVPGPRGKPIQILPAG......   98
Lcorn 18104 AgSp1.1 glycine rm1 1          GI................NPYYPMRPGR........................................................   87
Ncruc 58532and13532 AgSp1 glycine rm 1     ..........PGFNFG.......................................TPATTPTPIQGPGGKPIQVVPAG......  103
Mlaby 7048 AgSp1 glycine rm1a 1             ..........................................................TTTTKVKGPQGKPVLVVPAG......   52
Mhut 1271 AgSp1 glycine rm1 1               RTK........................................................RPSY.THVKDKPIQVIPAG......   96
Aarg AgSp1 FC511 glycine rm1 1             ......................PEEPSGRPMYPFGPGSPYGPGE.............QTTTTPIPGPDGKPLQIEPAG......  114
Atri AgSp1 genomic glycine rm1 1            ...................PKTPQGSDGQPMYPFGPGSPYGPGE.............QTTTTPIPGPDGKPLPIEPAG......  114
Msag 18528 AgSp1.2 glycine rm 1            ....................................................................................   16
Varen 6869 AgSp1.2 glycine rm1 1           ..............................................................PETHPAGRPIQIVPAG......   86
Varen 6869 AgSp1.1 glycine rm1 1           GKM....................................FGPM.........RPRRQRTTPPSVEGPNGKPIQIEPAG......   92
Amarm 50219 AgSp1.1 glycine rm1 2          GG.......................................................QTTPTPVPGPR................   88
Ncruc 58532and13532 AgSp1 glycine rm 2     ..........PGFNFG.......................................TPATTPTPIPGPGGKPIQVVPAG......  103
Mlaby 7048 AgSp1 glycine rm1a 2             ............FYFP.......V...............................TPKPTPTPVTGPGGKPIQIIPAG......  101
Mhut 1271 AgSp1 glycine rm1 2               ..........LI..................................................LGKASHGRPIQIIPAG......   92
Aarg AgSp1 FC511 glycine rm1 2             GGM.............PLSPYYPQGPGGQPMNPFAPGSPYGPGE.............QTTTTPIPGPDGKPLQIEPAG......  129
Atri AgSp1 genomic glycine rm1 2            GGM.............PLSPYSPQGPGGQPMYPFGPGSPYGPGE.............QTTTTPIP...................  116
Varen 6869 AgSp1.2 glycine rm1 2           ........................................QGMN.............FATTTPIPGPGGRPIQIEPAG......   99
Varen 6869 AgSp1.1 glycine rm1 2           GGY..P.................MRPR.........................................................   83
Amarm 50219 AgSp1.1 glycine rm1 3          ....................................................................GKPIQILPAG......   10
Ncruc 58532and13532 AgSp1 glycine rm 3     ..........PGFNFG.......................................TPATTPTPIPGPGGKPIQVVPAG......  103
Mhut 1271 AgSp1 glycine rm1 3               ..........PGFYFG...........................................MPQYIPGPGGRPIQIVPAG......   99
Aarg AgSp1 FC511 glycine rm1 3             ....................................................................................   22
Varen 6869 AgSp1.2 glycine rm1 3           ....................................................................................   11
Ncruc 58532and13532 AgSp1 glycine rm 4     ..........PGFNFG.......................................TPATTPTPIPGPGGKPIQVVPAG......  103
Mlaby 7048 AgSp1 glycine rm1a 4             ......................TGPGGQPLNPQGPISPFGPGGQSMNPLGPGGSGGQTTTPEPIPGPNGKPLQIVPAG......  128
Mhut 1271 AgSp1 glycine rm1 4               ..........PGFYFG...........................................MPQYIPGPGGRPIQIVPAG......   99
Ncruc 58532and13532 AgSp1 glycine rm 5     ..........PGFNFG.......................................TPATTPTPIPGPGGKPIQVVPAG......  103
Mhut 1271 AgSp1 glycine rm1 5               ..........PGFYFG...........................................MPQYIPGPGGKPIQIEPAG......   96
Ncruc 58532and13532 AgSp1 glycine rm 6     ....................................................................................   59
Mlaby 7048 AgSp1 glycine rm1a 6             ....................................................................................   46
Mhut 1271 AgSp1 glycine rm1 6               ..........PGFYFG...........................................MPQYIPGPGGKPIQIEPAG......   99
Mhut 1271 AgSp1 glycine rm1 7               ..........PGFYFG...........................................MPQNIPGPDGKPIQIVPAG......   99
Amarm 50219 AgSp1.1 glycine rm1 8          ..........PGFNFATP.......................................ATTPTPIPGPGGKPIQVIPAG......  103
Mhut 1271 AgSp1 glycine rm1 8               ....................................................................................   45
Varen 6869 AgSp1.2 glycine rm1 10          .................................................GSDEQGMNFVTTTPIPGPGGRPIQIEPAG......   37
Varen 6869 AgSp1.1 glycine rm1 11          .........................PG...GYPMGPGNPFRPMGP.........GRQRTTPSSVRGPKGKPIQIEPAG......   45
Varen 6869 AgSp1.2 glycine rm1 11          ........................................QGMN.............FATTTPIPGPGGRPIQIEPAG......   99
Varen 6869 AgSp1.2 glycine rm1 12          ........................................QGMN.............FATTTPIPGPGGRPIQIEPAG......   99
Aarg AgSp1 FC511 glycine rm1 13            DGQ.............PMYPSGP.............Q...GPGG.............QTTTTPIPGPDGKPIQIEPAG......   91
Varen 6869 AgSp1.2 glycine rm1 13          ........................................QGIN.............FATTTPIPGPGGRPIQIEPAG......   99
Varen 6869 AgSp1.2 glycine rm1 14          ....................................................................................   45
Mlaby 7048 AgSp1 glycine rm1a 15            GTT.......PG.........VVTGPGGEPLRQV..........HVRPAGPGTTP.GAKNPLPVTQSDGQPIQIVPAGPGTTPG  130
Aarg AgSp1 FC511 glycine rm1 15            ....................................................................................   62
Msag 18528 AgSp1.2 glycine rm 16           ........................GPQGPGGYPLGPGGQ........T....................................   16
Lcorn 18104 AgSp1.1 glycine rm1 37         GL................NPYYPMGPEGKPMYPYGPGSPFGPQGP.........GG.QTTKTPVPGPDGKPIQIVPAG......  126
Lcorn 18104 AgSp1.1 glycine rm1 38         GQ...........................PL......FPFGPQGP.........VG.QTTTTPIPGPDGKPIQIVPAG......  110
Lcorn 18104 AgSp1.1 glycine rm1 39         ....................................................................................   45
Atri AgSp1 genomic glycine rm1 47           .................................................................GPDGKPLQIEPAG......   13
Msag 18528 AgSp1.2 glycine rm 112          .......................RGPQGPGGYPLGPGSPFGPKGP.........GGQTTTPASVQGPDGEPIQVEPAG......   46
Msag 18528 AgSp1.2 glycine rm 113          G.............................GYPLGPGSPFGPKGP.........GGQTTTPASVQGPDGEPIQVEPAG......  115
Msag 18528 AgSp1.2 glycine rm 114          ....................................................................................   70
Msag 18528 AgSp1.2 glycine rm 115          ....................................................................................   35
consensus                                                                                     *   * ** ****** ***
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Amarm 50219 AgSp1.1 glycine rm1 1	....	98
Lcorn 18104 AgSp1.1 glycine rm1 1	....	87
Ncruc 58532and13532 AgSp1 glycine rm 1	....	103
Mlaby 7048 AgSp1 glycine rm1a 1	....	52
Mhut 1271 AgSp1 glycine rm1 1	....	96
Aarg AgSp1 FC511 glycine rm1 1	....	114
Atri AgSp1 genomic glycine rm1 1	....	114
Msag 18528 AgSp1.2 glycine rm 1	....	16
Varen 6869 AgSp1.2 glycine rm1 1	....	86
Varen 6869 AgSp1.1 glycine rm1 1	....	92
Amarm 50219 AgSp1.1 glycine rm1 2	....	88
Ncruc 58532and13532 AgSp1 glycine rm 2	....	103
Mlaby 7048 AgSp1 glycine rm1a 2	....	101
Mhut 1271 AgSp1 glycine rm1 2	....	92
Aarg AgSp1 FC511 glycine rm1 2	....	129
Atri AgSp1 genomic glycine rm1 2	....	116
Varen 6869 AgSp1.2 glycine rm1 2	....	99
Varen 6869 AgSp1.1 glycine rm1 2	....	83
Amarm 50219 AgSp1.1 glycine rm1 3	....	10
Ncruc 58532and13532 AgSp1 glycine rm 3	....	103
Mhut 1271 AgSp1 glycine rm1 3	....	99
Aarg AgSp1 FC511 glycine rm1 3	....	22
Varen 6869 AgSp1.2 glycine rm1 3	....	11
Ncruc 58532and13532 AgSp1 glycine rm 4	....	103
Mlaby 7048 AgSp1 glycine rm1a 4	....	128
Mhut 1271 AgSp1 glycine rm1 4	....	99
Ncruc 58532and13532 AgSp1 glycine rm 5	....	103
Mhut 1271 AgSp1 glycine rm1 5	....	96
Ncruc 58532and13532 AgSp1 glycine rm 6	....	59
Mlaby 7048 AgSp1 glycine rm1a 6	....	46
Mhut 1271 AgSp1 glycine rm1 6	....	99
Mhut 1271 AgSp1 glycine rm1 7	....	99
Amarm 50219 AgSp1.1 glycine rm1 8	....	103
Mhut 1271 AgSp1 glycine rm1 8	....	45
Varen 6869 AgSp1.2 glycine rm1 10	....	37
Varen 6869 AgSp1.1 glycine rm1 11	....	45
Varen 6869 AgSp1.2 glycine rm1 11	....	99
Varen 6869 AgSp1.2 glycine rm1 12	....	99
Aarg AgSp1 FC511 glycine rm1 13	....	91
Varen 6869 AgSp1.2 glycine rm1 13	....	99
Varen 6869 AgSp1.2 glycine rm1 14	....	45
Mlaby 7048 AgSp1 glycine rm1a 15	VVTGPGGKPLRTTPGPGLQFPTGPGK	156
Aarg AgSp1 FC511 glycine rm1 15	....	62
Msag 18528 AgSp1.2 glycine rm 16	....	16
Lcorn 18104 AgSp1.1 glycine rm1 37	....	126
Lcorn 18104 AgSp1.1 glycine rm1 38	....	110
Lcorn 18104 AgSp1.1 glycine rm1 39	....	45
Atri AgSp1 genomic glycine rm1 47	....	13
Msag 18528 AgSp1.2 glycine rm 112	....	46
Msag 18528 AgSp1.2 glycine rm 113	....	115
Msag 18528 AgSp1.2 glycine rm 114	....	70
Msag 18528 AgSp1.2 glycine rm 115	....	35
consensus		

X non-conserved
X similar
X $\geq$ 50% conserved